

Talha Bilal

Senior Software Engineer & Technical Team Lead

Java | Spring Boot | PKI | Keycloak | Java Card | Cryptography

talha.at43@gmail.com

linkedin.com/in/talha-bilal-a74477154

github.com/talha-bilal

talha-bilal.github.io/portfolio

PROFESSIONAL SUMMARY

Senior engineer and technical lead with 4+ years building secure enterprise platforms on Java Spring Boot, PKI, and identity systems. Designs microservices for certificate lifecycle, digital signing, and HSM integration (AWS CloudHSM, Utimaco, SoftHSM2). Experienced in end-to-end Keycloak integration with custom SPI development, Java Card applet design, Spring Security, REST/OpenAPI, Bouncy Castle, and PKCS#11.

CORE SKILLS

Java 21/17, Spring Boot, Spring Security, Microservices, REST, OpenAPI, PKI, PKCS#11, Keycloak, OAuth 2.0, OIDC, Keycloak SPI, Java Card, GlobalPlatform, Bouncy Castle, IAIK, S/MIME, AWS CloudHSM, SoftHSM2, Docker, Jenkins, RabbitMQ, Redis, PostgreSQL, MySQL

EXPERIENCE

Dictalabs - Technical Team Lead (Java Spring Boot & PKI)

Aug 2024 - Present

- Lead team delivering microservices PKI and cryptographic systems.
- Architect HSM services for key management, signatures, and RFC 3161 timestamping.
- Own PKCS#11 integrations (AWS CloudHSM, SoftHSM2); improve signing reliability.
- Led complete Keycloak integration with custom SPI providers for platform auth flows.
- Develop Java Card applets for secure on-card keys and cryptographic operations.

Codegic - Senior Software Engineer (Java Spring Boot & PKI)

Aug 2021 - Jul 2024

- Built CA, OCSP/CRL, and certificate lifecycle management.
- Developed S/MIME/CMS signing and Crypto-Core microservices with HSM integration.
- Delivered Keycloak SSO/OIDC and custom SPI extensions for internal services.
- Implemented Java Card applets and host apps for smart-card issuance workflows.
- Implemented CI/CD, Docker, and performance tuning for production crypto services.

SELECTED PROJECTS

- Keycloak IAM Integration - realms, OIDC clients, custom SPI providers
- Java Card Applets - secure on-card crypto and credential operations
- PKI Server - CA lifecycle, OCSP/CRL, HSM-backed issuance
- Secure Email Signing - S/MIME/CMS with RSA & ECDSA
- Workflow Management - PDF signing, approvals, audit trails
- Timestamp Authority - RFC 3161 compliant TSA
- PKCS#11 Service - REST layer for enterprise HSMs
- Crypto-Core - Key generation, CSR signing, tenant isolation

EDUCATION

B.S. Computer Science - KFUEIT, 2021